



9/1/2026

N + - Network Fundamentals & Building Networks.

Switch and Router Configuration Basics.

TASK 1.

- ❖ Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

➤ Creating VLANs in Cisco Packet Tracer:

Configuration:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Support
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# end
```

Result:

```
🔗 VLAN 10 → Sales
```

```
🔗 VLAN 20 → Support
```

Both VLANs were successfully created on the switch.

TASK 2.

- ❖ **Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command, Use the interface range command to select multiple ports efficiently.**

➤ **Assigning Ports to VLANs:**

Configuration

Assign 3 ports to VLAN 10 (Sales):

Switch> enable

Switch# configure terminal

Switch(config)# interface range fa0/1-3

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config-if-range)# exit

Assign 2 ports to VLAN 20 (Support):

Switch(config)# interface range fa0/4-5

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config-if-range)# exit

Verification:

```
Switch(config)# end
```

```
Switch# show vlan brief
```

Expected VLAN Membership:

VLAN	Name	Assigned Ports
10	Sales	Fa0/1, Fa0/2, Fa0/3
20	Support	Fa0/4, Fa0/5

The *show vlan brief* command confirms that the ports are assigned to the correct VLANs.

TASK 3.

❖ **Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.**

➤ Static Routing Between Two Routers:

Router 1 Configuration:

```
Router1> enable
```

```
Router1# configure terminal
```

```
Router1(config)# interface gigabitEthernet0/0
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

```
Router1(config)# interface gigabitEthernet0/1
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

Router 2 Configuration:

```
Router2> enable
Router2# configure terminal
Router2(config)# interface gigabitEthernet0/0
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit

Router2(config)# interface gigabitEthernet0/1
Router2(config-if)# ip address 192.168.2.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
```

Static Route on Router 1

The LAN behind Router 2 is **192.168.2.0/24**, so Router 1 is configured with:

```
Router1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Result: Router 1 can now reach the **192.168.2.0/24** LAN through Router 2 at **10.0.0.2**.

TASK 4.

❖ Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication, Think about router-on-a-stick or Layer 3 switch concepts.

➤ Inter-VLAN Communication:

Ping Test

A device in the **Sales VLAN (VLAN 10)** tries to ping a device in the **Support VLAN (VLAN 20)**.

Sales PC: 192.168.10.2

Support PC: 192.168.20.2

C:\> ping 192.168.20.2

Request timed out.

Why Does the Ping Fail?

The ping fails because **VLAN 10** and **VLAN 20** are **separate logical networks**. A Layer 2 switch cannot forward traffic directly between different VLANs.

Configuration to Allow Communication

Router on a Stick can be used by creating subinterfaces on the router, one for each VLAN:

```
Router(config)# interface g0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config)# interface g0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

The switch port connected to the router should be configured as a trunk port. This allows the router to route traffic between **Sales (VLAN 10)** and **Support (VLAN 20)**.